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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Sixth Semester B.Tech (CBCS) Degree Examinations

May 2025

EC2103-1 – INTERNET OF THINGS
(For EC)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. Which of the following is the standard for WiMax?
A) 802.16
B) 802.3
C) 802.15
D) 802.11
2. Which of the following is used to capture data from the physical world in IoT devices?
A) Sensors
B) Actuators
C) Microprocessors
D) Microcontrollers
3. Which of the following is not an actuator in IoT?
A) Stepper motor
B) A fan
C) An LED
D) Arduino
4. What is the component of an IoT system that executes a program?
A) A sensor
B) A microcontroller
C) An actuator
D) A digital to analog converter
5. ZigBee operates at frequency of
A) 2 GHz
B) 3.4 GHz
C) 2.4 GHz
D) 4.2 GHz
6. Full form of I2C is
A) inter-integrated circuit
B) inter-integrated connection
C) integrated-intelligent circuit
D) integrated-inter circuit
7. IoT gateway must provide _____
A) Protocol abstraction
B) Data storage
C) Security with hardware
D) Simple and fast installation
8. The huge number of devices connected to the Internet of Things has to communicate automatically, not via humans. What is this called?
A) Skynet
B) Bot 2 Bot
C) Machine 2 Machine
D) Intercloud
9. What does CoAP stand for in the context of web connectivity protocols?
A) Common Object Access Protocol
B) Constrained Application Protocol
C) Communication over Application Protocol
D) Connected Object Authentication Protocol
10. What does REST stand for in the context of web connectivity principles?
A) Representational State Transfer
B) Remote Service Transmission
C) Real-time Event Streaming
D) Resource Endpoint Security Transfer
11. Which protocol provides a lightweight and efficient alternative to HTTP for RESTful communication in IoT applications?
A) MQTT
B) CoAP
C) SOAP
D) TCP

12. What is the primary role of Media Access Control (MAC) in internet connectivity principles?
 A) Assigning IP addresses B) Managing network congestion
 C) Controlling access to network resources D) Ensuring data security
13. Which protocol is commonly used for transmitting email messages over the internet?
 A) FTP B) SMTP
 C) HTTP D) Telnet
14. IPv6 has an address length of
 A) 32 bits B) 64 bits
 C) 256 bits D) 128 bits
15. Hyper Text Transfer Protocol uses _____ port number
 A) 78 B) 80
 C) 82 D) 88
16. _____ reserves the port number according to specific application layer protocols.
 A) IANA B) ISO
 C) IETF D) ISRO
17. Relative humidity sensor senses using:
 A) Magnetic flux changes B) Resistance of air varying with moisture
 C) Piezoelectric effect in air D) Capacitance change between the parallel plates due to water vapours in air
18. Analog to Digital Converter resolution is expressed in _____
 A) Bytes B) Bits
 C) Word D) Nibble
19. Gyroscope is used to measure _____
 A) Angular velocity and linear acceleration B) Linear Acceleration
 C) Linear velocity D) Angular velocity
20. What is ESP8266?
 A) Sensor B) USB cable
 C) WIFI module D) Board

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Write about any 4 popular IoT boards in the development of IoT projects.	8	L*1	1	1
	b) Differentiate between IoT and M2M.	8	L2	2	1
2.	a) Explain the Oracle IoT architecture with a suitable diagram.	8	L1	1	1
	b) How are IoT Level1 being different from IoT Level2. Discuss with necessary diagrams.	8	L2	2	1
3.	a) Summarize the characteristics of IoT.	8	L2	1	1
	b) Discuss the architectural constraints of REST.	8	L1	2	1
Unit – II					
4.	a) List the features of Constrained Application Protocol.	8	L1	3	1
	b) Illustrate the actions of four layers of TCP/IP suite in internet communication.	8	L2	4	1
5.	a) Discuss CoAP-SMS protocol used in message communication.	8	L2	3	1
	b) Draw a diagram showing IPv4 header and explain its fields.	8	L1	4	1
6.	a) Discuss the data exchanges between HTTP web-objects.	8	L1	3	1
	b) With a neat diagram showing data link layer and adaptation layer protocols, discuss the 6LoWPAN in an IPv6 network.	8	L2	4	1

Unit – III

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|----|----|--|---|----|---|---|
| 7. | a) | With a neat diagram discuss the Architecture of vehicular IoT. | 8 | L2 | 5 | 1 |
| | b) | Explain the E-Health IOT application in detail. | 8 | L1 | 5 | 1 |
| 8. | a) | Discuss the design elements of home automation using IoT.
Mention its advantages. | 8 | L1 | 5 | 1 |
| | b) | Explain the working of Smart Cards as an IoT application. | 8 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
